

Predictive Modeling of Student Behavior in Virtual Reality: A Machine Learning Approach

Abstract: In the ever-evolving landscape of education, Virtual Reality (VR) environments offer a promising avenue for enhancing student engagement and learning experiences. However, understanding and predicting student behavior within these immersive settings remain challenging tasks. Recent studies have described learners' perceptions and post-hoc outcomes, but none have offered practical models that predict behavior as it unfolds. This paper presents a comprehensive study on the predictive modeling of student behavior in VR using machine learning techniques. This study used a rich dataset of student interactions, movements, and task outcomes from a VR orientation program and applied machine learning techniques to forecast student engagement and success. We developed regression and classification models using feature engineering and model selection to reveal hidden behavioral patterns. The best regression model predicted a continuous engagement score with high accuracy (cross-validated $R^2 > 0.95$ and hold-out R^2 approx 0.98), and a multinomial classification model correctly categorized engagement level with 84.9% accuracy (kappa = 0.77) on test data. These results demonstrate the feasibility of real-time predictive analytics in VR. By uncovering the relationship between student behavior and VR interactions, this work provides valuable insights for educators, instructional designers, and developers seeking to optimize virtual learning environments.

Introduction

Virtual Reality (VR) has become a prominent instructional medium because its immersive, three-dimensional spaces can heighten learners' sense of presence, interactivity, and motivation (Bower & Jong 2020 ; Craig & Kay 2023 ; Ferguson et al. 2025). Across age groups and disciplines, studies consistently report significant gains in engagement and skill acquisition when students practice concepts in richly simulated VR scenarios (Checa et al. 2023 ; Liu & Shirley 2021). The proliferation of VR applications is evident across the entire educational spectrum, from enhancing foundational concepts in elementary schools to providing specialized training in higher education and professional development (Maryani et al. 2024; Checa et al. 2023). This widespread adoption is fueled by VR's unique capacity to construct highly interactive, engaging,

and contextually rich learning environments (Chen & Syu 2024). Such platforms can simulate scenarios and offer experiences that would be impractical, hazardous, or prohibitively expensive to replicate in traditional classroom settings (Checa et al. 2023). The value of this technology has been particularly highlighted in the context of distance and online learning, which have become increasingly prevalent (Sadanala et al. 2024; Li et al. 2022). VR offers a potent solution to the persistent challenges of student isolation and disengagement that often characterize remote educational formats, providing a shared virtual space that can foster a sense of presence and community (Valenti et al. 2020; Li et al. 2022).

Machine-learning (ML) techniques have already proven effective at forecasting academic performance and engagement in conventional online platforms by mining click-stream, assessment, and forum data (Jawad et al. 2022 ; Nguyen et al. 2016 ; Yan et al. 2025).

Early-warning dashboards derived from such models allow instructors to target support before students disengage or fail, thereby improving retention and personalisation (Jawad et al. 2022 ; Yan et al. 2025). Transferring this predictive power into immersive VR is appealing because VR generates even richer behavioural traces ranging from spatial paths and eye-gaze to voice interactions yet research offering real-time VR behaviour prediction remains scarce (Sadanala et al. 2023 ; Ferguson et al. 2025).

Accordingly, the present study introduces a detailed VR-orientation dataset that logs successes, failures, exploration patterns, and social exchanges for each learner, then applies classification and regression algorithms to forecast engagement level and task success during the session itself. By demonstrating accurate, moment-by-moment prediction, we aim to bridge the gap between immersive design and adaptive support, enabling educators to personalise VR experiences and intervene precisely when students need help.

Literature Review

1. Educational benefits and challenges of VR

Systematic reviews show that immersive VR reliably boosts motivation, self-efficacy, and learning outcomes when activities are authentic and well scaffolded (Bower & Jong 2020 ; Craig & Kay 2023). A recent meta-analysis further confirms that desktop VR can effectively enhance

academic achievement (Liu et al. 2024). Comprehensive systematic reviews of the literature reveal a sustained and growing interest in immersive VR within the context of higher education, indicating that the technology is now considered a mainstream tool for post-secondary instruction (Craig & Kay 2023). This trend is not confined to universities; scientometric analyses also point to a clear and accelerating increase in VR-related research within elementary education, demonstrating its perceived value across different developmental stages (Maryani et al. 2024). A particularly powerful driver of this expansion has been the global shift towards remote and online instruction, where VR is increasingly leveraged as a means to enhance and improve the quality of distance learning (Sadanala et al. 2024; Li et al. 2022). Domain-specific experiments echo these findings; for example, a VR computer-assembly game improved autonomous problem-solving skills among undergraduates (Checa et al. 2023), other studies show how VR-supported interaction develops technology-related learning (Lind, 2024), and a “virtual study-abroad” programme sustained intercultural learning during pandemic travel bans (Liu & Shirley 2021). In these contexts, VR is seen as a vital technology for bridging the physical distance between learners, fostering a sense of co-presence and community that is often lacking in conventional web-based courses (Li et al. 2022; Valenti et al. 2020).

Despite these advantages, researchers caution that excessive sensory input can impose cognitive load, and ill-fitting headsets may induce fatigue or motion sickness, thereby dampening learning (Zhang et al. 2024 ; Ferguson et al. 2025). Accessibility and inclusion issues also persist; a scientometric review notes gender-related barriers in VR science activities and calls for inclusive design guidelines (Pyrkosz-Pacyna et al. 2024).

2. Machine-learning predictions in conventional e-learning

In virtual-learning environments outside VR, supervised ML models routinely achieve high accuracy when predicting final grades, dropout risk, or behavioural engagement using early-semester interaction data (Jawad et al. 2022; Wang & Yu, 2025). Random-forest classifiers, for instance, distinguish pass–fail outcomes even with imbalanced datasets by employing data-balancing techniques (Jawad et al. 2022), while deep-learning architectures fuse multimodal signals like text, video, biometrics to estimate engagement with promising F-scores (Yan et al.

2025). Meta-analyses confirm that such predictive dashboards help instructors intervene early, thus improving retention and personalisation (Nguyen et al. 2016 ; Yan et al. 2025).

While predictive analytics is becoming commonplace in conventional online learning platforms, its application within immersive VR educational environments remains a nascent and comparatively underdeveloped area of research. The unique characteristics of VR data which is inherently spatial, continuous, multimodal, and captures a far wider range of interactive behaviors present both new challenges and unprecedented opportunities for predictive modeling (Kreß et al. 2024). The spatial dimension alone, tracking a user's position and orientation in 3D space, adds a layer of complexity not found in traditional clickstream data (Kreß et al. 2024). The literature contains numerous calls for the development of practical, validated models that can harness this rich data stream for the purpose of real-time pedagogical intervention, yet very few studies to date have successfully delivered on this promise. This represents a significant frontier for learning analytics research.

3. Current knowledge on modelling behaviour inside VR

Research focusing on post-experience VR metrics reveals mixed links between perceived engagement and learning gain; for example, Sadanala et al. (2024) found no significant correlation between engagement questionnaire scores and orientation-quiz marks, suggesting conventional metrics may miss critical in-session behaviours (Sadanala et al. 2024). Qualitative analyses likewise document exploratory hesitancy, ergonomic discomfort, and distraction but stop short of predictive modelling (Sadanala et al. 2023 ; Ferguson et al. 2025). Calls for real-time analytics emphasize leveraging VR-specific signals such as gaze, head rotation, and object-interaction frequency to flag confusion or disengagement before performance suffers (Checa et al. 2023 ; Kee et al. 2024). Nevertheless, a tangible, publicly documented pipeline that captures those signals and validates predictions against learning outcomes has yet to appear highlighting a clear research gap our study addresses (Sadanala et al. 2024).

Methodology

Dataset

The data was generated within a bespoke educational VR environment built on the OpenSimulator (OpenSim) platform. This semi-immersive desktop VR system replicates a portion of a real university campus and was designed to provide an accessible and scalable orientation experience for new online students. The choice of a desktop platform over more immersive Head-Mounted Displays (HMDs) was a deliberate design consideration to minimize potential adverse effects such as motion sickness and to lower the financial and technical barriers for student participation.(Park & Lee, 2020)

Within this environment, students were tasked with completing five distinct, activity-based modules designed to familiarize them with university resources, faculty, and program information. These activities included:

- Interacting with Non-Player Character (NPC) versions of professors to learn about their research.
- Locating and learning about specific technology resources, such as an eye-tracker.
- Inquiring about funding opportunities through virtual conversations and interactive displays.
- Participating in a "Scavenger Hunt" to find learning resources.
- Completing an "Activity Board" game to gain helpful learning tips. The student plays a game named 'Activity Board' with 10 learning tasks.

Participants

The dataset comprises behavioral data from 115 participants recruited via convenience sampling over four academic semesters, beginning in Fall 2019. Participants were new undergraduate and graduate students enrolled in a College of Education. The sample included 50% females and 27% males and 23% preferred not to disclose.

Variable	Frequency	Percentage
Age		
18-23	22	19.1%
24-30	21	18.3%
31-40	27	23.5%
41-50	12	10.4%
50 or older	8	7%
Not revealed	25	21.7%
Gender		
Female	58	50.4%
Male	31	27%
Not revealed	26	22.6%
Computer skills		
Advanced (can handle daily use and most problems)	42	36.5%
Beginner (may need help during daily use)	9	7.8%
Intermediate (can handle daily use but may need help for some problems)	39	33.9%
Not revealed	25	21.7%
Experience in a 3DVRLE or a 3D game		
High (I am a frequent user of 3D VRLE or player in 3D games.	4	3.5%
Intermediate (I have some experience in 3D VRLE or 3D games)	27	23.5%
Low (I have little experience in 3D VRLE or I'm not a gamer of 3D games)	59	51.3%
Not revealed	25	21.7%

Figure. 1 Demographic characteristics of participants.

Data Collection and Feature Extraction

Prior to commencing the study, Institutional Review Board (IRB) approval for data collection was obtained. The participating students were informed of the study procedures and provided their consent before the actual intervention. The raw data then was captured through comprehensive screen recordings of each participant's entire orientation session. These recordings were subsequently processed and coded using the Behavioral Observation Research

Interactive Software (BORIS), a tool for systematic event logging. Trained coders used the software to view the recordings and meticulously annotate specific, observable behaviors related to task engagement and performance, generating a timestamped log for each action. This coding process produced the rich, quantitative dataset foundational to this study. The final dataset contains over 100 distinct behavioral features for each student, capturing a detailed telemetry of their in-VR actions.

Data Pre-Processing

It was necessary to perform data preprocessing before applying predictive modeling.

The data preprocessing includes dealing with inconsistent data, eliminating data noise and imputing missing values through a variety of techniques and strategies (Jawad et al. 2022). The process involved a multi-stage filtering approach. First, any feature column with more than 30% missing values was removed, as these variables were too sparse to provide reliable predictive information. Following this, any participant or row who was still missing more than 30% of their data across the remaining features were also excluded from the analysis.

To handle the few remaining missing (NA) values without further reducing the sample size, median imputation was employed. This standard technique preserves the integrity of the dataset while ensuring it is complete and ready for the modeling algorithms, which cannot handle missing values. Finally, non-predictive identifier columns (e.g., `participant_id`, `name` etc.) were removed to prevent data leakage. The resulting final model dataframe served as the clean and complete dataset for all subsequent regression and classification analyses.

Results

Following the pre-processing stage, a comprehensive modeling strategy was employed to predict student success. A two modeling strategy was employed to predict student success:

1. **Regression Analysis:** To predict the continuous engagement score, which represents the total time a student spent successfully completing tasks.

2. **Classification Analysis:** To predict a student's categorical engagement level. The engagement score variable was converted into a three-level factor (Not Engaged, Engaged, Highly Engaged) based on terciles (33rd and 66th percentiles).

Regression Analysis: Predicting engagement

To identify the most influential predictors of student success, an initial feature selection was performed using a Random Forest model. The model's variable importance plot (varImpPlot) highlighted key behaviors, such as time spent reading chats and navigating the environment, as strong initial predictors as shown in figure 2.

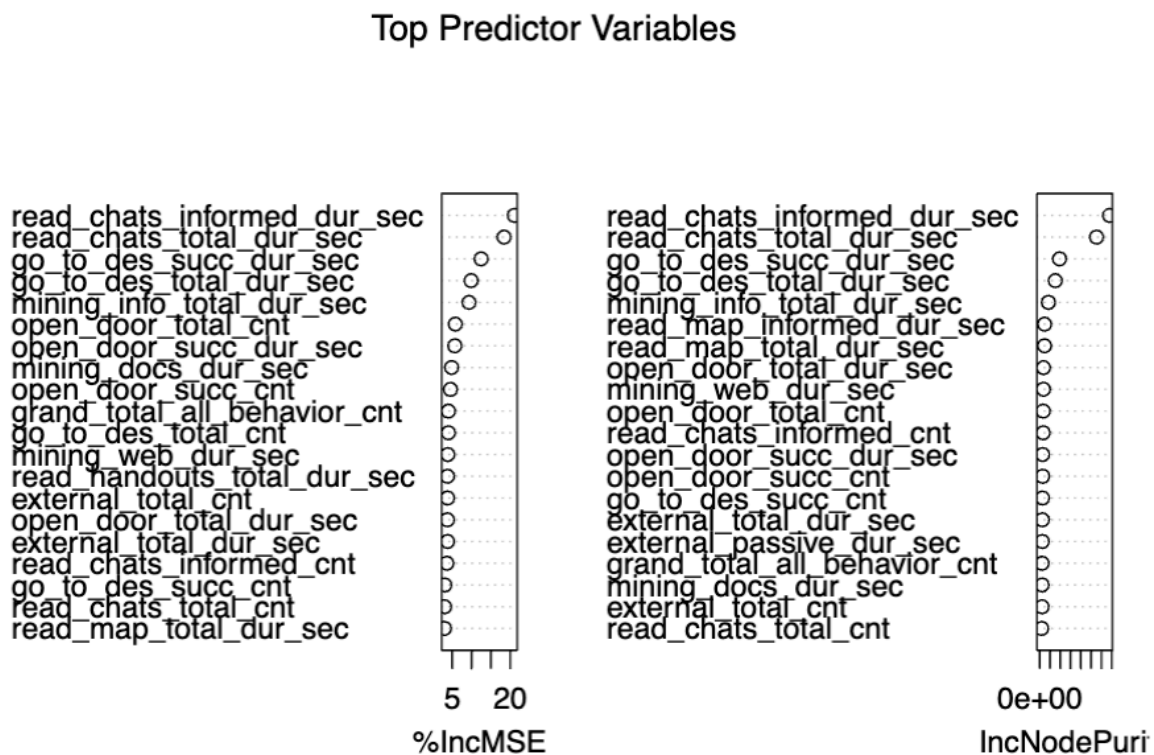


Figure. 2 Variable Imp Plot for Top Predictors variables

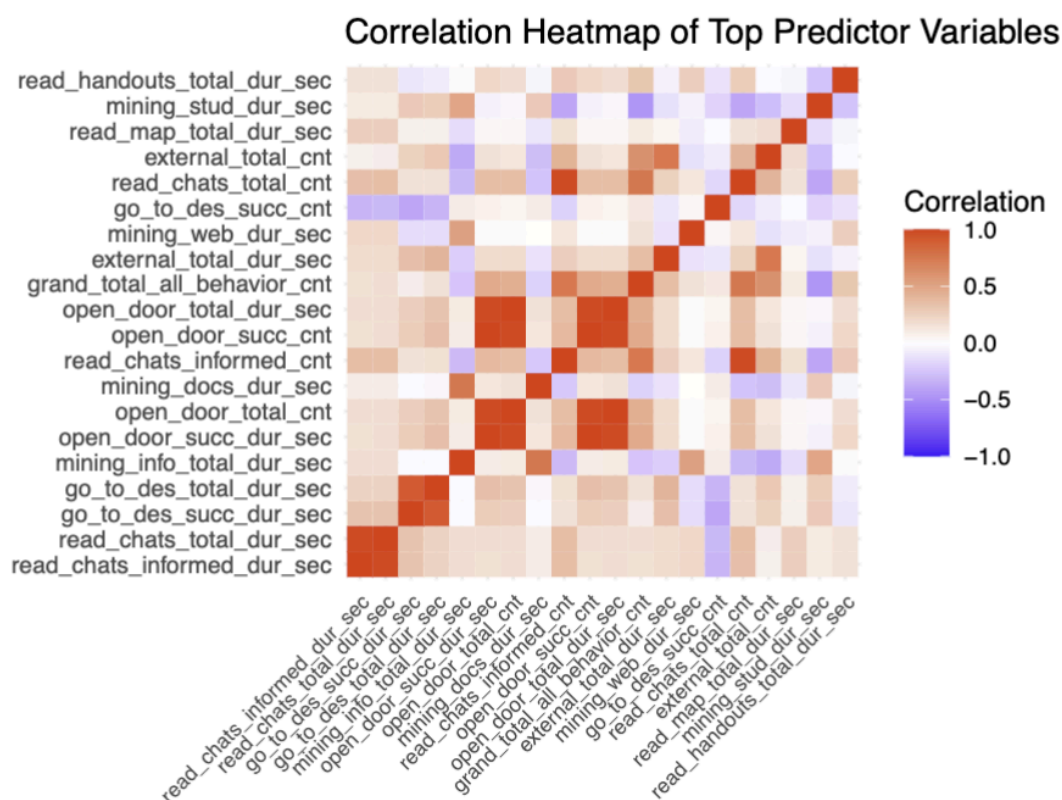


Figure 3. Correlation heatmap of top predictors

As we can see in figure 3., correlation heatmap of these top predictors revealed several highly correlated features, indicating the need for a more parsimonious model to avoid multicollinearity. Based on the importance plot and correlation analysis, a reduced set of the most impactful and less correlated features were selected (figure 4. and figure 5).

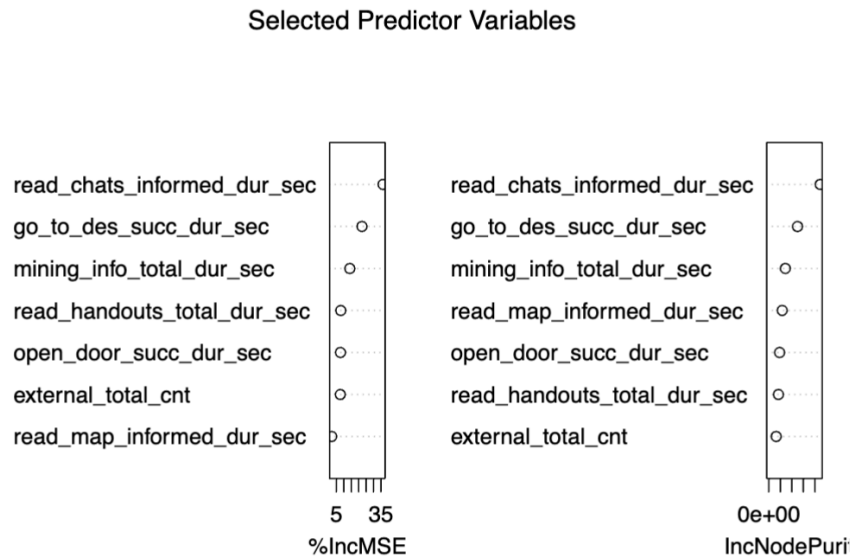


Figure 4. Variable Importance plot for selected predictors

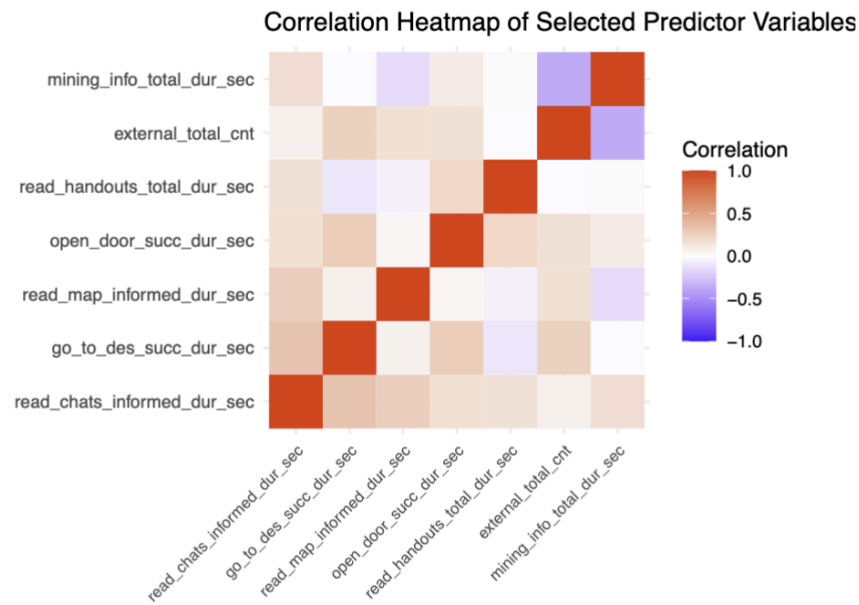


Figure 5. Correlation heatmap of selected predictors

A suite of regression models was trained using 5-fold cross-validation on this reduced feature set. The SVR-Linear and Bayesian Ridge models were selected as top performers, both achieving a mean R^2 above 0.95 and the lowest Root Mean Square Error (RMSE) values as displayed in figure 6.

Table 1: Cross-validated performance metrics for regression models

Model	Mean R^2	SD R^2	Mean RMSE (s)	SD RMSE (s)	Mean MAE (s)	SD MAE (s)
Linear	0.958	0.017	325.194	146.275	228.511	68.996
Lasso	0.955	0.023	335.391	168.682	231.022	75.503
Bayesian Ridge	0.959	0.015	321.433	138.594	226.930	66.664
Random Forest	0.854	0.103	685.975	324.360	515.013	174.454
GBM	0.835	0.140	605.368	364.593	433.777	176.555
XGBoost	0.844	0.092	604.393	282.974	464.862	164.970
SVR Linear	0.967	0.009	264.867	31.467	201.845	26.332
SVR Radial	0.834	0.126	641.754	379.215	412.009	151.837
SVR Poly	0.573	0.215	409574.908	912677.438	91169.100	202242.580

Figure 6. Cross-validated performance metrics for regression models

The two best models were then evaluated on a final 70/30 hold-out split, representing completely unseen data. The Bayesian Ridge model slightly outperformed the SVR-Linear model on the test set, achieving an exceptional R^2 of 0.975 as depicted in figure 7.

Table 2: Final regression performance on the hold-out test set

Model	Final R^2	Final RMSE (s)	Final MAE (s)
Bayesian Ridge	0.975	262.414	195.989
SVR-Linear	0.915	772.455	346.076

Figure 7. Final regression performance on the hold-out set

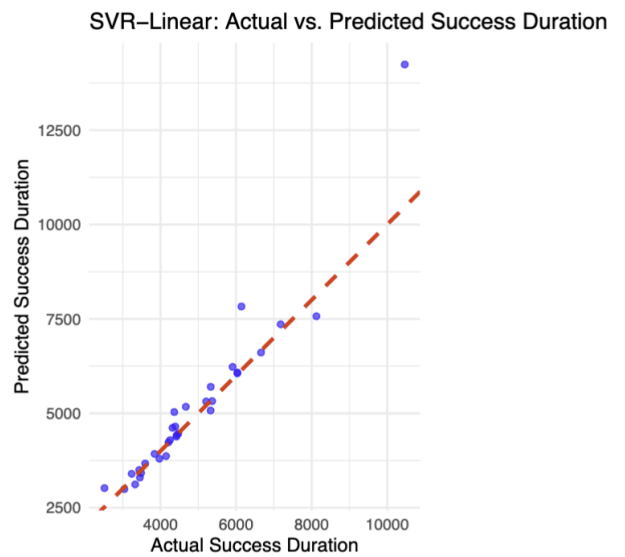
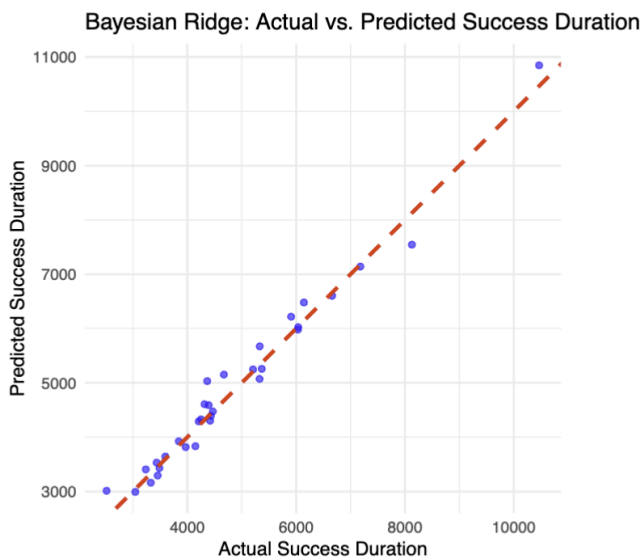


Figure 8. Actual vs. Predicted success duration plot

Classification Analysis: Predicting Engagement Level

The secondary analysis aimed to classify students into one of three engagement levels. The distribution of these levels was relatively balanced, providing a solid foundation for modeling as shown in figure 9.

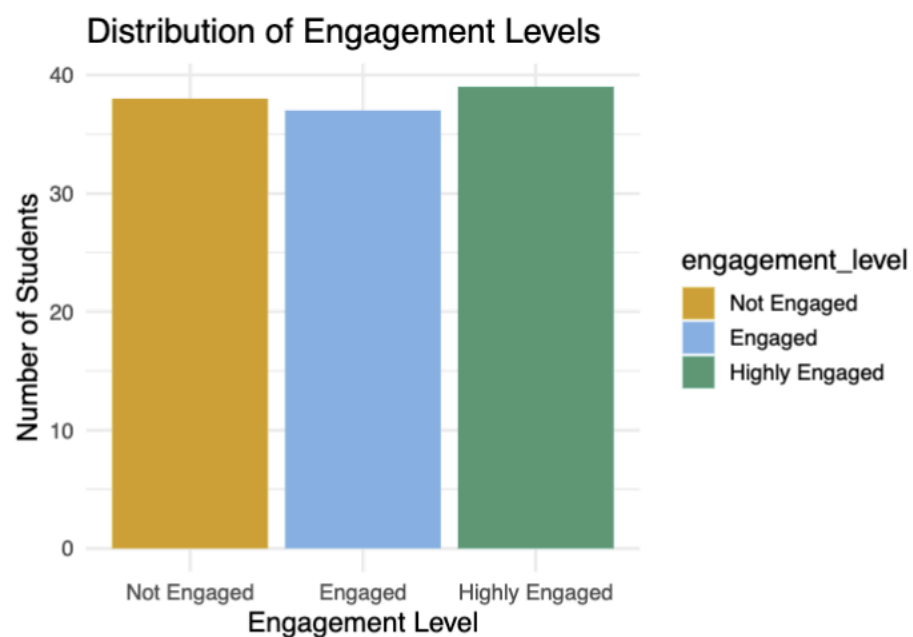


Figure 9. Distribution of engagement levels

Using the same set of predictors as regression models, three models were trained and evaluated via 10-fold cross-validation: Multinomial Logistic Regression, Support Vector Machine (SVM), and Random Forest. The Multinomial Regression model demonstrated the best performance, with a mean accuracy of 0.831 and a mean Kappa of 0.744 as displayed in figure 10.

Table 3: Model Performance Using Optimal RFE Features

Model	Mean Accuracy	SD Accuracy	Mean Kappa
Multinomial Regression (RFE)	0.831	0.107	0.744
SVM (RFE)	0.804	0.121	0.705
Random Forest (RFE)	0.748	0.111	0.622

Figure 10. Model performance using top predictors

To assess final performance on unseen data, the models were trained on a 70% partition of the data and tested on the remaining 30%. The confusion matrices for the Random Forest and Multinomial Regression models are shown in the figure 12 below. The Multinomial Regression model achieved a final accuracy of 84.9% and a Kappa of 0.773 on the hold-out test set, correctly identifying 10 of the 11 "Highly Engaged" students.

Table 4: Final performance on the hold-out test set

Model	Accuracy (95% CI)	Kappa	Balanced Accuracy	Macro Precision	Macro Recall	Macro F1
Random Forest	0.758	0.636	0.818	0.772	0.758	0.762
Multinomial Regression	0.848	0.773	0.886	0.848	0.848	0.848

Figure 11. Final performance on the hold-out test for classification models

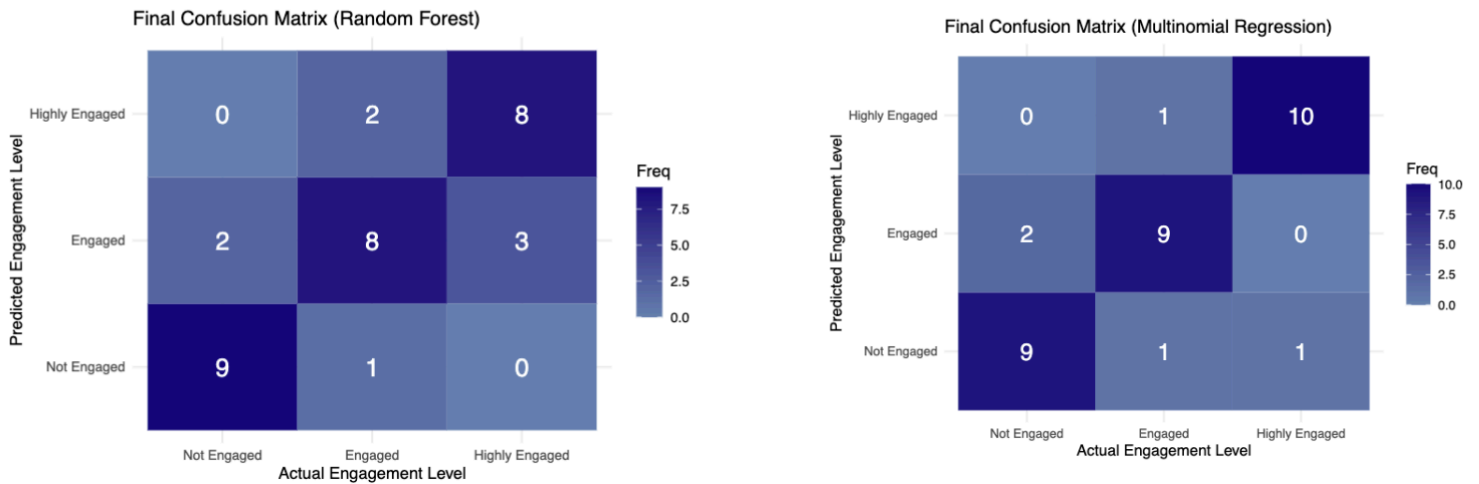


Figure 12. Confusion matrix for classification models

Discussion

Comparison with the real video data to confirm the results

To further investigate the behaviors identified by our predictive models, we conducted an in-depth qualitative analysis of a two-hour video recording of one participant. This analysis revealed the practical manifestations of our key predictor variables. For instance, the high importance of *read_chats_informed_dur_sec* was vividly illustrated when the participant, after a period of struggle, deliberately paused to read informational chat logs, which then directly led to the successful completion of the task. This confirms that this variable is not just a statistical artifact but a representation of a crucial, observable learning strategy.

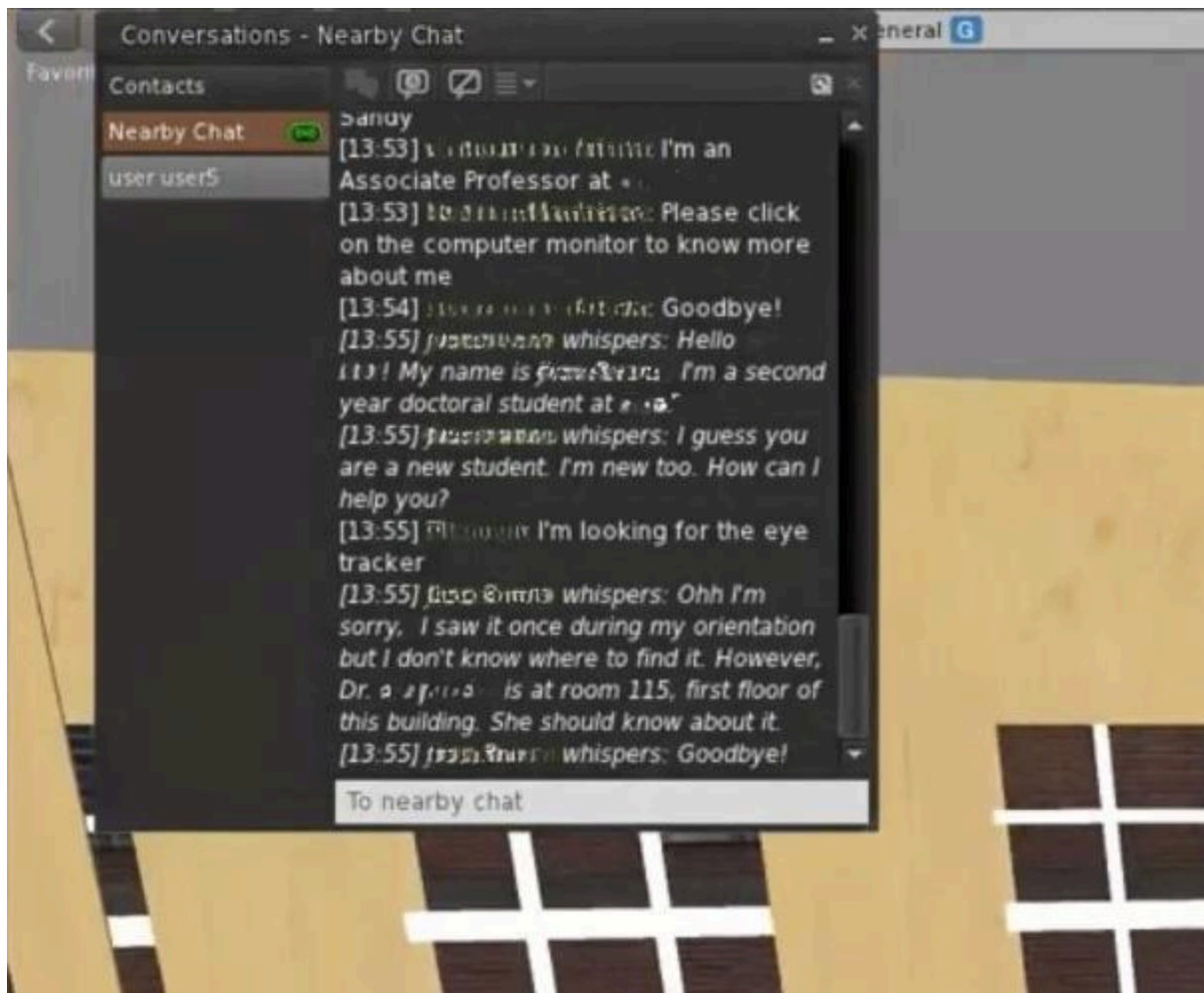


Figure 13. Analysis example of *read_chats_informed_dur_sec*

Furthermore, the video analysis showed us how a few key things worked together. We noticed a clear pattern: the participant would look for information, like reading a map or gathering details at a display, and then immediately use that information to successfully navigate. This pattern stood in stark contrast to periods of inactivity or struggle, which were characterized by a lack of such information-seeking actions. The participant's actions were methodical, showcasing a student who was actively orienting themselves, gathering necessary information, and then acting on that information to progress. This helped us confirm that our model accurately identified

effective problem-solving in the virtual environment: a cycle of asking questions and taking deliberate action. This triangulation of quantitative and qualitative data strengthens our conclusion that in-world behavioral telemetry can serve as a robust indicator of student engagement and success.

Limitations

While this study provides a robust proof-of-concept, it is important to acknowledge its limitations, which in turn point toward promising avenues for future research. First, the study was conducted using a non-immersive desktop VR system. Future work should seek to replicate these findings in fully immersive VR using HMDs to determine if the identified behavioral signature of engagement holds true in environments with a higher sense of presence. Second, the dataset was derived from a convenience sample of students from a single college, which may limit the generalizability of the findings. Replicating the study with more diverse student populations is a necessary next step. Third, the data collection relied on a labor-intensive manual coding process using BORIS software.

Conclusion

This study demonstrates that machine learning models can accurately predict student engagement and success in a VR learning environment by leveraging in-world behavioral data. Using a dataset of over 100 behavioral features logged from a virtual orientation program, we developed regression and classification models that achieved high predictive performance, for instance explaining ~97% of variance in engagement time and ~85% accuracy in classifying engagement level. The key behaviors identified such as information seeking and goal-directed actions provide actionable insights into what effective learners do in VR. These findings pave the way for more adaptive and responsive virtual learning experiences. In practice, educators and VR system designers can use our approach to implement real-time support interventions, ensuring that students remain engaged and receive help at the moment it is needed. By bridging immersive educational design with predictive analytics, this work contributes to more personalized and data-informed learning environments.

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